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Q1 A smart healthcare system processes large volumes of patient data such as temperature readings,

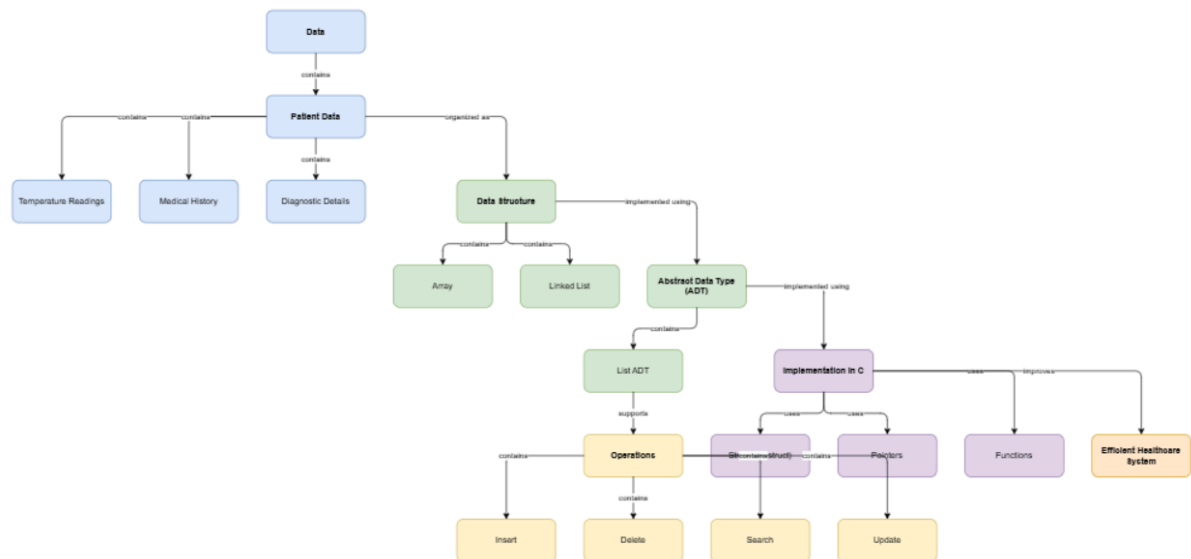
medical history, and diagnostic details. Construct a concept map linking the progression from Data →

Data Object → Data Structure → Abstract Data Type (ADT) → Implementation in C. Identify relevant

sub-concepts such as structures, pointers, and operations, and establish meaningful relationships among

them. Organize the concept map hierarchically and justify how each level contributes to efficient data

organization, processing, and improved system performance.



Q21 A student attendance system records daily presence data. Construct a concept map linking Array →

Fixed Size → Sequential Access → Linked List → Dynamic Allocation → Flexibility. Explain how each

structure works and justify which is more suitable for dynamic data updates.

